

HARI KRISHNAA N

Embedded Systems & Electronics Builder | Hardware Maker
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PROFILE

Hands-on maker and embedded systems enthusiast who builds things because it sounds cool. Passionate about electronics, firmware, and IoT — from burning Arduinos at midnight to shipping PCB-based wearables and real-time telemetry systems. Founder of HYNEX, where I prototype hardware-integrated AI solutions. Pursuing B.E. in Mechanical & Mechatronics (Additive Manufacturing); actively exploring PCB designing alongside firmware and 3D manufacturing.

TECHNICAL SKILLS

Embedded & IoT	Arduino, ESP8266/ESP32, PCB Design, Sensors, Circuit Wiring, Electronics Assembly
Firmware & Code	Python, C, JavaScript, TypeScript — Embedded Scripting, REST APIs, MQTT
PLC & Industrial	Siemens S7 (Basic) — Ladder Logic fundamentals, PLC programming concepts
3D & AM	FDM · SLA · SLS · Product Design · Additive Manufacturing · Process Optimisation
IoT Platforms	ThingSpeak, Firebase, AWS IoT, MQTT, Git, Postman
AI & Automation	Generative AI, RAG, Computer Vision, NLP, Multi-Agent LLM, Prompt Engineering
Web & Mobile	MERN Stack, React Native, Flutter, Vite, REST APIs

HARDWARE & EMBEDDED PROJECTS

- GoKart Telemetry & 3D Simulation System** | Arduino · ESP8266 · Python · ThingSpeak · 3D Viz
Designed and assembled a sensor array (speed, temperature, RPM, G-force, pressure) on a go-kart chassis. ESP8266 streams live data to a cloud dashboard; built a 3D simulation of kart dynamics for performance analysis and fault detection.
- FitSync — Smart Wearable Health Monitor** | ESP8266 · SpO2/HR Sensors · Flutter · Firebase
Built a wearable IoT device monitoring SpO2, heart rate, body temperature, and motion. Handled PCB layout, sensor calibration, and wireless data transmission to a Flutter app with real-time vitals, trend analytics, and health alerts.
- Industrial Air Quality & Leak Detection Node** | Arduino · Pressure/Humidity Sensors · Python · MQTT
Developed a distributed sensor node network monitoring compressed air pressure, humidity, and flow rate. Integrated ML-based anomaly detection with live dashboard alerts for predictive maintenance.
- Lunar Hazard AI — Computer Vision Safety System** | Python · OpenCV · Deep Learning
AI-powered hazard detection for autonomous navigation: identifies surface anomalies, classifies risk zones, and generates safe-path recommendations using computer vision and deep learning.
- FlowMind AI** | Python · Multi-Agent LLM
Autonomous workflow engine converting meeting inputs into structured tasks with smart escalation using multi-agent LLM architecture — hardware-software bridge system.

EXPERIENCE & TRAINING

- Founder & Lead Developer — HYNEX** 2024 – 2026
 - Delivered end-to-end hardware and software products — sensor integration, PCB testing, 3D-printed enclosures, firmware, and client deployment.
 - Built IoT sensor networks, automation workflows, and agentic AI systems for client solutions.
- Manufacturing Intern — Sree Ram Industries** 25 Days | 2025
 - Live factory floor: production workflow monitoring, quality inspection, mechanical tolerances, and QC processes.

Electronics Intern — Salzer Industrial Controls

Internship | 2024

- ▶ Circuit wiring, PCB testing, quality inspection — hands-on with industrial-grade control panels and relay circuits.

Web Development Intern — NullClass

1 Month

- ▶ Full-stack apps (React · Node · MongoDB); REST API integration; real code-review and sprint workflows.

Generative AI Training — Internshala

6 Weeks

- ▶ Prompt engineering, LLM app development, RAG pipelines on practical datasets.

RESEARCH WORKS

- ▶ Waste-to-Watt: Municipal solid waste & biomass → electrical energy via thermochemical conversion.
- ▶ Waste-to-Value (Fuel): Non-recyclable plastics → synthetic fuel via optimised pyrolysis.
- ▶ Waste-to-Value (Filament): Closed-loop post-consumer plastics → affordable FDM-ready 3D printing filament.
- ▶ Multi-Process AM: FDM/SLA/SLS process optimisation for stronger, lighter, sustainable printed components.

EDUCATION

B.E. Mechanical & Mechatronics (Additive Manufacturing) — SNS College of Engineering, Coimbatore 2023 – Present

Higher Secondary — SRT Universal Higher Secondary School, Sathyamangalam

2018 – 2023

CERTIFICATIONS & ACHIEVEMENTS

Certifications: AWS Academy AI/ML Intermediate • Azure AI Fundamentals (Microsoft) • Coursera Technical Foundations & Cloud • Python Fundamentals (Scaler) • MS Excel Fundamentals (Microsoft) • GeeksforGeeks Programming Logic & DSA • Preplnsta DSA & Aptitude

Achievements: 1st Place — Ideathon & 24-Hour Hackathon in KPR College 2025 & 26 • Finalist — Smart India Hackathon • 24-Hour Hackathon Participant (Multiple Colleges) • National Symposium Paper Presenter

Languages: English (Fluent) • Tamil (Fluent) • German (Basic)